

DANA 4830 - FINAL EXAM Summer 2026

Context:

While Rosati et al., (2024) concentrated on reviewing differential gene expression (DGE) analysis pipelines for identifying specific biomarkers, Syed et al. (2024) employed Machine Learning (ML) approaches to identify gene biomarkers and molecular signatures. **Given your expertise in both bioinformatics and data science, this task requires you to critically evaluate the differences between these two approaches, with a focus on optimizing the pipeline through computational reduction and minimal background knowledge.**

Task 1: Compare and Contrast Biomarker Identification Approaches

- Task 1.1: Review both articles and highlight the key differences between the two approaches used to identify biomarkers. Please write in a clear, concise manner using your own language, and limit your response to one page (approximately 300 words). **Note that I will not read and mark this part if I see the writing is being done by a LLM.** [25 point]
- Task 1.2: Provide 2 diagrams with the following requirements. [25 points]
 - Diagram 1 illustrates a comprehensive pipeline outlining the analytical workflow used by bioinformatics professionals to identify molecular signatures with potential biomarker significance. This diagram is directly related to the article published by Rosati et al. in 2024. Please include all relevant techniques and key concepts mentioned in the article, such as differential gene expression (from sections 2.1-2.4), pathway enrichment analysis, ROC curve analysis, and survival statistical analysis.
 - Diagram 2 demonstrated your pipeline that you conducted in your mid-term 1.

Task 2: Demonstration

- Task 2.1: Conduct a biomarker identification analysis using bioinformatics tools and omics techniques on the GEO 75214 dataset building upon your existing knowledge from Midterm 1. As a hint, align this analysis with the concepts presented in the article by Rosati et al. (2024). Upon identifying potential biomarkers, compare the results of this approach with those obtained during Midterm 1. To address any balance issues that may arise, employ the same techniques used in your previous teamwork, which will require re-running specific aspects of your earlier work. [25 points]
- Task 2.2: Retrieve and merge four existing datasets: GSE6791, GSE9750, GSE63514, and GSE67522, into a unified master dataset. Next, validate both pipelines conducted earlier on this combined dataset. To identify differentially expressed genes (DEGs), employ the following cut-off criteria: $|\log_{2}FC| \geq 1.5$ and adjusted P-value < 0.05 . Please note that these datasets were previously analyzed by Wang et al. in their 2022 study. However, keep in mind that sample sizes may differ due to updates in omics data platforms. [25 points]

Requirement of submission

Submit your work using the same structure and template as in your mid-term 1. **Please ensure that your submission allows me to replicate your work accurately.** If I can't replicate, I might be able to give marks on the work you did. Note that some parts of your analysis may involve R programming language. In this case, please export the names of any R packages used in R-Studio, and/or provide instructions or a README file for integrating these packages under Python notebooks.

References:

Rosati, D., Palmieri, M., Brunelli, G., Morrione, A., Iannelli, F., Frullanti, E. and Giordano, A. (2024). Differential gene expression analysis pipelines and bioinformatic tools for the identification of specific biomarkers: A Review. *Computational and Structural Biotechnology Journal*, 23.
doi:<https://doi.org/10.1016/j.csbj.2024.02.018>.

Syed, A.H., Abujabal, H.A.S., Ahmad, S., Malebary, S.J. and Alromema, N. (2024). Advances in Inflammatory Bowel Disease Diagnostics: Machine Learning and Genomic Profiling Reveal Key Biomarkers for Early Detection. *Diagnostics*, [online] 14(11), p.1182.
doi:<https://doi.org/10.3390/diagnostics14111182>.

Wang, D., Liu, Y., Cheng, S. and Liu, G. (2022). Identification of Novel Genes and Associated Drugs in Cervical Cancer by Bioinformatics Methods. *Medical Science Monitor*, 28.
doi:<https://doi.org/10.12659/msm.934799>.